

Kevin Schoenholzer Postdoctoral Researcher Università della Svizzera italiana (USI) Lugano, Switzerland kevinschoenholzer@gmail.com <https://kevinschoenholzer.com>

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Editor-in-Chief *Journal of Statistics and Data Science Education* Taylor & Francis

Re: Submission of manuscript “Empiria: A Modular-Synthesizer Platform for Simulation-Based Statistics Education”

Dear Editor,

I am pleased to submit the enclosed manuscript for consideration as an Article in the *Journal of Statistics and Data Science Education*.

The paper describes the **Methods** plugin of **Empiria**, an open-source (GPL-3.0) suite for VCV Rack 2 — a free modular-synthesizer environment that I have repurposed as a live, signal-flow teaching surface for statistics. Methods comprises fourteen modules that together expose a full inferential pipeline — data-generating process → sample-window measurement → estimation → inference → diagnostics → reporting — as patch cables on a rack. The paper sets the design in the context of two decades of GAISE-aligned calls for simulation-based reasoning [Cobb 2007; GAISE 2016; Tintle et al. 2014], the constructionist heritage of Papert and Resnick, and the older tradition of physical statistical instruments (the Galton board, the analog computer, the chemical-plant trainer).

The contribution sits squarely within JSDSE’s scope. The plugin spans the four competencies the journal regularly addresses — sampling-distribution intuition, simulation-based inference, regression, and reproducible computational workflows — and is designed for use in classrooms ranging from high-school general education to graduate research-methods seminars. The implementation draws on the established numerical literature (Lentz continued fractions for the incomplete beta in the t -test [Press et al. 2007]; Efron’s bias-corrected and accelerated bootstrap [Efron 1987]; Sanger’s rule for online PCA [Sanger 1989]) so that the underlying numerics are auditable rather than implicit. To keep the contribution focused, the paper describes five **anchor modules** (*Sample, Frame, Regress, Test, Boot*) that carry the simulation-based-inference argument in depth, and summarises the remaining nine Methods modules in a compact reference table that points to the project manual for full detail.

Empiria’s broader suite — four additional plugins covering agent-based social models, network epidemiology, spatial dynamics, and behavioral-economics models — shares the Methods plugin’s panel grammar, patch-cable conventions, and deterministic-seed reproducibility story. To keep the JSDSE submission narrowly focused on statistics education, those plugins are noted only briefly here (with one cross-plugin Schelling demonstration in Workflow A to illustrate that the same signal-flow approach generalises) and are the subject of a separate companion paper for the computational social-science community.

Three worked examples in the paper illustrate the range of use cases: (a) a visceral Schelling-segregation demonstration suitable for introductory teaching; (b) a fully seeded, reproducible

two-sample t -test pipeline; and (c) an empirical verification of a closed-form probability (four-coin-flip Bernoulli runs) that makes the case for methodological robustness explicit. A dedicated “Methodological robustness” section documents the numerical algorithms used and their references, addressing the legitimate concern that an unconventional environment might trade rigor for appeal.

The manuscript is approximately 8,500 words. All software, panels, documentation, and three starter patches generated by an included Python script are available under GPL-3.0 at the URL listed on the title page. The paper has not been previously published and is not under consideration elsewhere. I am the sole author and have no conflicts of interest to declare.

I have suggested five reviewers (listed separately) with expertise spanning statistics education, computational social science, simulation-based inference, and open-source educational software. I would be grateful for the opportunity to have this work assessed by the JSDSE community, whose readership most closely matches the audience I have built *Empiria* for.

Thank you for your consideration.

Sincerely,

Kevin Schoenholzer